5G – The Second Phase (3GPP Rel.16 and 17)

Course objectives:

During 2019, the 5G Mobile Communication System became a reality with the 3GPP releasing the first set of Release 15 Technical Specifications. Following that, network elements and mobile devices became available for commercial deployment, enabling network operators to introduce the new technology into their networks.

In 3GPP's terms, Release 15 refers to the very first phase of 5G, predominantly designed to address data rate enhancements for enhanced Mobile Broadband (eMBB).

The 3GPP Release 16, however, aims at adding the remaining functionalities to make 5G fully compliant with the strict requirements of ITU's IMT-2020.

In this training course we'll explore the variety of key features to enable the remaining two pillars of IMT-2020, namely massive Machine Type Communication (mMTC) and Ultra-Reliable Low Latency Communication (URLLC).
We shall further discuss, how other organizations (e.g. GSMA and NGMN) contribute to assure interoperability between 5G networks for fluent user experience.

The participant will learn about Release 16 enhancements pertaining network topologies, Radio-, Core- and Transport Networks aspects.
Plenty of use cases are provided to illustrate how to apply the Release 16 features in a real-life scenario, before taking a glimpse at further additions by Release 17 and the future 6G.

Duration Instructional methods:

Option 1: On-Site Training: 4 days,
Classroom-style lecture using Power-point slides with interactive discussion,
Questions & Answers after each module.

Option 2: Virtual Classroom: 4 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 8 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

Target Audience:

The course is addressed to technical staff, manager and executive level from the technology, marketing and management field, regulators and government officials, executives, requiring an overview about the 5G Release 16 features (The 5G Second Phase).

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Course Contents:

- Introduction
  - Refresher: ITU IMT-2020 Requirements
  - 5G Building Blocks: eMBB, mMTC and URLLC and their requirements
  - The 5G Phases
  - Standardization: 3GPP Rel.16, Rel.17 and 6G
  - NGMN contribution:
    - User Experience
    - Enhanced Services
    - Mobile Network Operators
    - Consumer Requirements
    - Vertical Requirements (e.g. SME Business, Transport and Traffic, Health Care, VR/AR, Aviation and Drones etc.)

- 3GPP Rel. 16 System Architecture and Functionality
  - Rel.16 Enhancements
    - LTE in Rel.16
    - 5G in Rel.16
    - Fixed-Mobile Convergence
    - Control Plane / User Plane Separation of EPC Nodes
    - Identifiers
    - Multicast/Broadcast in 5G
  - 5G Network Architecture in Rel.16
    - Roaming and Non-Roaming Architectures
    - Interworking with Non-3GPP Networks
    - Edge Computing
  - Dual Connectivity
    - Multi-radio Dual Connectivity with EPC
    - Multi-radio Dual Connectivity with 5GC
  - NG-RAN Architecture
  - IMS in 5G

- 3GPP Rel.16 Radio Network and User Equipment
  - Key Release 16 Enhancements
  - Radio Network
    - 5G MIMO and Adaptive Antennas
    - 5G Radio Access
    - 5G gNB Functions
    - 3GPP RAN Interfaces
    - The Split Architecture of RAN
    - IAB
    - 5G Network Layers

- Cloud RAN
- 5G Spectrum and advances in 5G Frequencies
- Radio Aspects
  - Bandwidth
  - SUL
  - Dynamic Spectrum Sharing
  - 5G Antennas
  - Radio Performance
  - OFDM in Release 16
  - Modulation and Coding
  - Frame Structure
  - 5G Channels
  - General Protocol Architecture
  - Physical Layer Procedures
  - Physical Layer Measurements
  - Quality of Service

- Core- and Transport Network Aspects
  - 5G Core Network Services
  - Network Functions Virtualization
    - SDN
    - NFV
  - 5G Cloud Architecture Concept
    - Data Center as a Base for 5G Architecture
    - Network as a Service
  - 5G Release 15 and 16 Network Functions
    - NFs Enhanced in Rel.16
    - NF's Added in Rel.16
    - NF Discovery and Network Slicing
  - The Transport Network
  - IMS for 4G and 5G Voice service
    - IMS Architecture
    - Roaming
    - VoLTE Infrastructure Options
    - Circuit-Switched Fallback
    - Single Radio Voice Call Continuity
    - Interworking in 4G/5G
    - Requirements for IMS Voice

- Release 16 Features and Use Cases
  - The 3GPP Use Cases of SMARTER
  - Enhancement of Ultra-Reliable Low Latency Communications
  - 5GS Enhanced Support of Vertical and LAN Services
  - Advanced V2X Support

- Satellite Access in 5G
- Wireless and Wireline Convergence Enhancement
- Location-Based Services
- Mission Critical Services
- Public Warning System
- Streaming and TV
- Cloud-Based Functions and Edge
- Virtual Reality, Augmented Reality, and Extended Reality
- SON
- Support for Energy Saving
- Enablers for Network Automation Architecture for 5G
- 5G Voice
- Sidelink
- Verticals Support
- Non-public Networks

- 3GPP Release 17 and Beyond
  - Drones (Unmanned Aerial System)
  - MBMS
  - Machine Learning and Artificial Intelligence
  - Use Cases of 6G

- 5G Security Aspects
  - 5G Security Architecture
  - Enhanced 5G Security
  - Network Security Procedures
    - Keys in 5G
    - 5G Identifiers
    - Network Key Storage and Procedures
    - 5G Key Derivation
    - Security Aspects of Network Slicing
  - SIM in the 5G Era
    - UICC Profiles in 5G
    - Changes in Authentication
    - SIM Evolution
    - eSIM
    - eSIM Architecture
    - Security Certification of 5G SIM and Subscription Management
  - Security Certification of Data Centers
  - GSMA Security Controls

Pre-requisites:

A solid understanding of 3GPP Mobile Communication Systems and a working knowledge of 3GPP Release 15 aspects

Course Code: 5GRel16